

Spurious level repulsion in geophysical random-matrix analysis: declustering, seasonal gating, and the Wishart null

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Abstract

The consecutive spacing ratio $\langle r \rangle$ is increasingly used to claim Wigner–Dyson (GOE) “level repulsion”, and hence organisation, in geophysical and climate data. But $\langle r \rangle$ is two-ended, and several mundane data structures mimic GOE without any underlying repulsion. We document three such mechanisms with controlled demonstrations and give the correct controls. (1) *Declustering*: removing points closer than a fraction of the median spacing drives even pure Poisson points to GOE-like $\langle r \rangle$, manufacturing regularity from randomness. (2) *Seasonal gating*: when events can occur only in a fixed season, the inter-event interval distribution is bimodal and the naive $\langle r \rangle$ lands between Poisson and GOE as an artifact of that marginal. For 41 major sudden stratospheric warmings the naive interval $\langle r \rangle = 0.48$ is permutation-invariant (scalar), while the correct de-seasonalised test — the per-winter event count — shows mild *under-dispersion* (variance/mean = 0.67, 95% CI excluding unity), i.e. a weak refractory clock, not level repulsion. (3) *The Wishart null*: the eigenvalue spacing of a sample correlation matrix is GOE for independent data, so “GOE level spacing” in a climate correlation matrix is the finite-sampling null; the physical signal lies in the eigenvalue *density* (departures from Marchenko–Pastur), not in its spacing. The honest diagnostics are raw $\langle r \rangle$ with the coefficient of variation, the permutation (shuffle) test, an appropriate physical null, and — for correlation matrices — the eigenvalue density rather than its spacing.

1 Introduction

Random-matrix theory (RMT) provides a celebrated test for “order” in a sequence of levels: the spacings between consecutive levels follow Poisson statistics for an uncorrelated sequence and the Wigner–Dyson statistics of the Gaussian ensembles for a spectrum with level repulsion (Mehta, 2004). A convenient, unfolding-free measure is the mean ratio of consecutive spacings (Oganesyan & Huse, 2007; Atas et al., 2013),

$$\langle r \rangle = \left\langle \frac{\min(s_i, s_{i+1})}{\max(s_i, s_{i+1})} \right\rangle, \quad s_i = x_{i+1} - x_i, \quad (1)$$

which takes the reference values 0.386 (Poisson), 0.531 (GOE) and 0.603 (GUE). The statistic is attractive precisely because it needs no unfolding, and it is now applied well beyond quantum spectra — to earthquake catalogues, eruption sequences, climate indices and ore-deposit records — often to argue that a geophysical system shows “level repulsion” and is therefore organised by a single coherent process.

This note is a caution. The ratio $\langle r \rangle$ is two-ended: a value above the Poisson reference can arise from genuine level repulsion, but equally from a narrow or structured marginal spacing distribution with no sequential repulsion at all, or from pre-processing. We document three distinct ways in which geophysical and climate analyses obtain GOE-like $\langle r \rangle$ without any underlying repulsion, each with a controlled demonstration, and we state the diagnostic that exposes it. The first mechanism was reported in a companion study of orogenic-gold vein spacing (Chen, 2026a); we recapitulate it and add two more that are specific to climate data, unifying all three under a single rule: *compare the observed statistic against the permutation of its own data and against the correct null, never against the textbook GOE value alone.*

2 The ratio statistic and the permutation test

Genuine level repulsion has two fingerprints: a marginal spacing distribution that is depleted at small spacings (so that $P(s \rightarrow 0) \rightarrow 0$), and sequential anti-correlation (spectral rigidity). The ratio $\langle r \rangle$ responds to both. The decisive control is the *permutation (shuffle) test*: randomly re-order the spacings, which preserves the marginal distribution exactly but destroys any sequential correlation, and recompute $\langle r \rangle$. If $\langle r \rangle_{\text{obs}} \approx \langle r \rangle_{\text{shuf}}$, the value is a property of the marginal alone — it carries no sequential repulsion, however far above Poisson it lies (Chen, 2026b). We also report the coefficient of variation $\text{CV} = \sigma_s / \bar{s}$, which is 1 for Poisson, < 1 for regular sets and > 1 for clustered sets. With these two numbers, and an appropriate null, each of the three hazards below is immediately diagnosable.

3 Hazard 1: declustering fabricates GOE

A common pre-processing step is *declustering*: removing points that lie closer together than some fraction of the median spacing. This manufactures regularity from any input. Figure 1a shows the effect on pure Poisson points: as the minimum-separation threshold increases from 0 to 0.7 median spacings, $\langle r \rangle$ rises monotonically from the Poisson value 0.386 through the GOE value 0.531 to 0.57. No repulsion is present in the data; the GOE-like value is entirely an artifact of the thinning rule. Mechanistically, declustering injects an *artificial hard core*: by deleting the smallest spacings it forces $P(s \rightarrow 0) \rightarrow 0$, the very depletion that defines level repulsion, so it manufactures a steric-exclusion signature the data do not contain — the inverse of the genuine hard cores documented in nucleosome arrays and ice-wedge networks. This hazard was identified, with a field example, in a study of orogenic-gold vein spacing (Chen, 2026a), where raw vein positions are clustered ($\langle r \rangle = 0.34$) but rise to 0.44–0.50 under declustering in lockstep with random controls. The remedy is simple: never decluster before computing a spacing statistic; report the raw $\langle r \rangle$ with CV.

4 Hazard 2: seasonal gating

Many geophysical events can occur only within a fixed window of the year — winter storms, monsoon bursts, stratospheric warmings. This seasonal gating imprints a strong periodic structure on the sequence of event times that has nothing to do with the system’s internal dynamics, yet contaminates the naive interval statistic.

We illustrate with major sudden stratospheric warmings (SSWs), the abrupt reversals of the winter polar vortex. We use the 41 central dates of major Northern-Hemisphere SSWs in the JRA-55 reanalysis (1958–2023) from the NOAA SSW Compendium (Butler et al., 2017; Charlton & Polvani, 2007). Computed naively on the calendar sequence, the inter-event interval ratio is

Table 1: Three routes to spurious GOE-like $\langle r \rangle$ and the diagnostic that exposes each.

Hazard	Spurious result	Correct diagnostic
Decustering	random points $\rightarrow \langle r \rangle \approx \text{GOE}$	raw $\langle r \rangle$, no thinning; CV
Seasonal gating	bimodal intervals $\rightarrow \langle r \rangle = 0.48$	permutation test; de-seasonalised counts
Wishart sampling	iid data $\rightarrow \text{bulk } \langle r \rangle \approx \text{GOE}$	eigenvalue density vs Marchenko–Pastur

$\langle r \rangle = 0.484$ — comfortably between Poisson and GOE, and easily mis-read as partial “repulsion”. It is nothing of the kind. The interval distribution is strongly bimodal (Fig. 1b): 85% of intervals exceed 270 days (the gaps across each SSW-free summer) and only a handful fall within a winter. The permutation test confirms the artifact: $\langle r \rangle_{\text{shuf}} = 0.475$, indistinguishable from the observed value, so the elevation above Poisson is a property of the bimodal, seasonally-imposed marginal — a scalar, not sequential repulsion. The intermediate value follows directly from the bimodality: consecutive cross-summer gaps (the dominant mode) give ratios near unity, while the occasional pairing of a within-winter interval with a cross-summer gap gives ratios near zero, and the average of these extremes falls between the Poisson and GOE references. Because this is a property of the marginal alone, the permutation test reproduces it exactly.

The physically meaningful question — does the vortex resist a second warming (a refractory period) or fire at random? — is answered only after de-seasonalising. Counting major SSWs per winter season over the 66 winters spanned (Fig. 2) gives a mean of 0.62 and a variance of 0.42, a variance-to-mean ratio of 0.67 with a bootstrap 95% confidence interval of $[0.49, 0.86]$ that excludes unity. The counts are therefore mildly *under-dispersed*: winters with two SSWs are rarer, and winters with exactly one are commoner, than a Poisson process would produce, consistent with a recovery (refractory) timescale after each event (Ineson et al., 2024). This is a weak scalar refractory clock — the opposite end of the spacing axis from clustering, but not Wigner–Dyson level repulsion, and invisible to the naive interval $\langle r \rangle$.

5 Hazard 3: the Wishart null

RMT is also applied to climate *correlation matrices*: the $P \times P$ matrix of correlations among P spatial cells or indices, whose eigenvalues are then examined for GOE level statistics, taken as evidence of collective organisation (Santhanam & Patra, 2001). Here the trap is more basic. The eigenvalue spacing of a sample correlation (Wishart) matrix is GOE *by construction*, independent of any real correlation in the data. Figure 1c demonstrates this: for P *independent* Gaussian time series of length T — zero true correlation — the bulk eigenvalue ratio of the sample correlation matrix is $\langle r \rangle \approx 0.53$ across a range of (P, T) , squarely at the GOE value. GOE-class level spacing is thus the finite-sampling *null* for any correlation matrix; reporting it as a climate finding is to report the null. The bulk spacing is fixed by random-matrix universality — the local eigenvalue statistics shared by essentially every large matrix in the ensemble, independent of the entries’ physical origin — so it tests that universality class, not any climate or geophysical dynamics.

The genuine RMT signal in a correlation matrix is not in the spacing but in the eigenvalue *density*: eigenvalues that escape the Marchenko–Pastur support expected for pure noise mark real collective modes (Marchenko & Pastur, 1967), exactly as established for financial covariance matrices (Laloux et al., 1999; Plerou et al., 2002). The level spacing of the bulk is GOE whether or not those modes exist. The remedy is to test the eigenvalue density against Marchenko–Pastur, and reserve any spacing analysis for the de-noised spectrum.

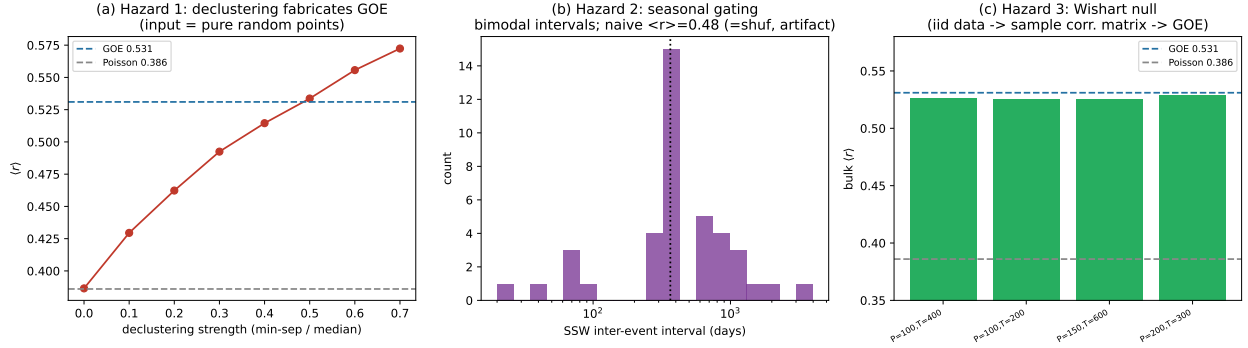


Figure 1: Three ways to fabricate GOE-like level repulsion. **(a)** Minimum-separation declustering drives pure Poisson points from $\langle r \rangle = 0.386$ through the GOE value to 0.57. **(b)** Sudden-stratospheric-warming inter-event intervals are bimodal (note the cross-summer gaps near and beyond one year); the naive $\langle r \rangle = 0.48$ equals its permutation value and is a seasonal artifact. **(c)** The bulk eigenvalue ratio of a sample correlation matrix built from *independent* Gaussian series sits at the GOE value for every (P, T) tested: GOE spacing is the Wishart null.

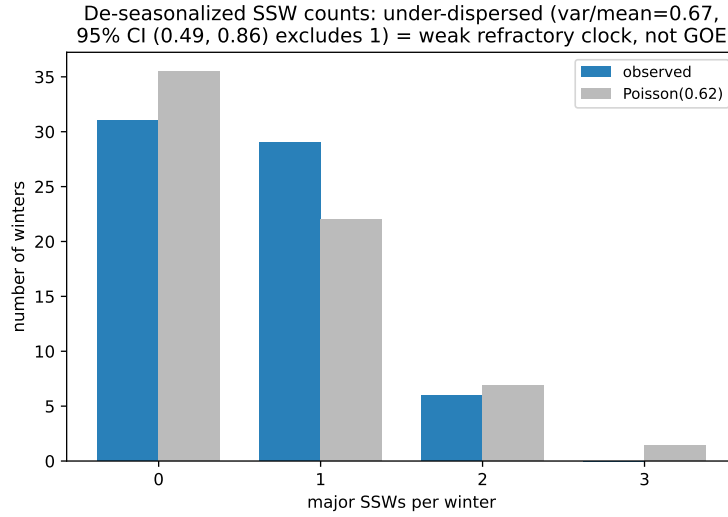


Figure 2: De-seasonalised sudden stratospheric warmings: the per-winter event count is mildly under-dispersed relative to Poisson (variance/mean = 0.67, 95% CI [0.49, 0.86], excluding unity) — a weak refractory clock, not level repulsion. Winters with exactly one SSW are commoner, and winters with two rarer, than Poisson predicts.

6 Correct practice

The three hazards share a single remedy, summarised in Table 1. Before interpreting any $\langle r \rangle$ above the Poisson value as level repulsion: (i) use the raw data, applying no declustering or thinning; (ii) run the permutation (shuffle) test — if $\langle r \rangle_{\text{obs}} \approx \langle r \rangle_{\text{shuf}}$ the value is a marginal (scalar) property, not sequential repulsion; (iii) report the coefficient of variation alongside $\langle r \rangle$, since clustered sets ($\text{CV} > 1$) can sit near or below Poisson while a narrow marginal ($\text{CV} < 1$) lifts $\langle r \rangle$ without repulsion; (iv) for periodically gated events, de-seasonalise (e.g. analyse per-season counts) before drawing dynamical conclusions; and (v) for correlation matrices, test the eigenvalue density against Marchenko–Pastur rather than reading GOE off the bulk spacing. Genuine Wigner–Dyson repulsion survives all of these; the spurious cases do not.

7 Conclusion

GOE-like values of the consecutive spacing ratio are easy to obtain by accident. Declustering manufactures them from random points; seasonal gating manufactures them from the calendar; and the eigenvalue spacing of any sample correlation matrix is GOE for independent data. None of these is level repulsion. The permutation test, the coefficient of variation, de-seasonalisation, and the Marchenko–Pastur density test distinguish a real Wigner–Dyson spectrum from its common impostors. Applied to sudden stratospheric warmings, the honest analysis finds not level repulsion but a weak refractory clock — a small, real, and correctly-attributed result.

Data and code availability

The SSW central dates are from the NOAA SSW Compendium (Butler et al., 2017) (<https://cs1.noaa.gov/groups/cs18/sswcompendium/>). All demonstrations are synthetic or use these public dates; code and derived tables are at <https://github.com/Ruqing1963/spurious-level-repulsion> and archived on Zenodo (DOI assigned on release).

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